

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I M.DEVI SRILATHA(192525236) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Cloud Architecture for Video Streaming Platform

A media company with millions of concurrent users needs a cloud architecture that provides high scalability, low latency, and minimum buffering.

Proposed Architecture

Users → CDN → Load Balancer → Auto-Scaling Servers → Streaming Storage

The video files can be stored in cloud object storage. A CDN can cache popular videos at edge locations close to users.

Scalability and Caching

- Cloud object storage can store a large number of video files.
- CDN caches frequently watched videos at edge locations.
- Users receive videos from a nearby CDN server.
- This reduces latency and load on the main servers.
- Auto-scaling can automatically increase the number of servers when traffic increases.
- Servers can be reduced when traffic becomes low.

Load Balancing

A load balancer distributes requests among multiple streaming servers.

For example:

Users → Load Balancer → Server 1 | Server 2 | Server 3 | Server 4

If one server becomes overloaded or fails, traffic can be sent to another healthy server.

Auto-Scaling

During popular events, the number of users can suddenly increase. Auto-scaling creates additional computing resources automatically.

During low-traffic periods, unnecessary resources are removed.

Cost Optimization

The company can reduce cloud costs by:

1. Using CDN caching to reduce origin-server traffic.
2. Using auto-scaling instead of keeping maximum servers running.
3. Using storage tiers for old or less frequently accessed videos.
4. Using reserved or committed resources for predictable workloads.

5. Monitoring resource usage and removing unused resources.
6. Compressing videos and using suitable streaming formats.

Additional Performance Enhancements:

7. Use adaptive bitrate streaming so video quality automatically adjusts to each user's network speed, reducing buffering.
8. Deploy edge computing nodes for tasks like transcoding closer to users, lowering latency for live content.
9. Use GPU-based transcoding to prepare multiple resolution formats quickly during traffic spikes.
10. Apply request coalescing at the CDN layer so repeated requests for the same segment do not overload origin servers.

Conclusion

Using CDN, cloud storage, load balancing, auto-scaling, and caching can support millions of users with less buffering. Cost can also be controlled by using resources according to actual demand.

2. Multi-Cloud Container Deployment for Fintech

A fintech startup using multiple cloud providers may face problems such as different configurations, deployment methods, networking, and security policies.

Proposed Solution

The company can use containerization with Kubernetes to provide a common deployment and management platform.

Application → Container → Kubernetes → Cloud A / Cloud B / Cloud C

The same container image can be tested and deployed across different cloud environments.

Container Orchestration

Kubernetes can manage containers automatically.

It can:

- Deploy applications.
- Restart failed containers.
- Scale containers.
- Perform load balancing.
- Manage application updates.
- Maintain the required number of application instances.

Service Discovery

Microservices need to communicate with each other. Kubernetes provides service discovery using services and internal DNS.

For example:

Payment Service → Order Service → Database Service

Applications can find services using their service names instead of depending on fixed IP addresses.

Scaling

Kubernetes can automatically increase or decrease the number of containers based on CPU usage, memory usage, or application traffic.

For example:

Low Traffic → 3 containers

High Traffic → 10 containers

Benefits of Multi-Cloud

- Reduces dependency on one cloud provider.
- Improves availability.
- Provides flexibility.
- Can help avoid vendor lock-in.
- Different providers can be selected based on their services and pricing.

Risks of Multi-Cloud

- More complex management.
- Network communication can be complicated.
- Different cloud services may have different features.
- Security policies must be managed carefully.
- Data transfer between clouds can increase cost.
- Troubleshooting can become difficult.

Security Considerations:

- Scan container images for vulnerabilities before deployment across any cloud provider.
- Use a centralized secrets-management service so credentials are never hard-coded into container images.
- Apply Kubernetes network policies to restrict which services can communicate with each other.
- Enforce consistent role-based access control (RBAC) policies across all connected clouds.

Conclusion

Using containers and Kubernetes provides consistency across cloud providers. Multi-cloud improves flexibility and availability, but it also increases management and security complexity.

3. Backup and Disaster Recovery for E-Learning Platform

An e-learning platform can lose important data because of accidental deletion, software errors, or system failures. A strong backup and disaster recovery strategy is therefore required.

Proposed Architecture

Application → Primary Database → Backup → Replication → Secondary Region

The platform should use cloud-native backup, database replication, and versioning.

Backup

Regular backups should be taken for:

- Student information.
- Course data.
- Examination records.
- Assignment files.
- Application databases.

Backups should be stored separately from the primary system.

Replication

Important data can be replicated to another availability zone or region.

If the primary system fails, the replicated data can be used to restore the application.

Versioning

Versioning keeps previous versions of files.

For example, if an administrator accidentally deletes a course document, an earlier version can be restored.

RTO and RPO

RTO (Recovery Time Objective):

It defines how quickly the application should be restored after a failure.

Example:

If RTO = **30 minutes**, the application should be restored within 30 minutes.

RPO (Recovery Point Objective):

It defines the maximum acceptable amount of data loss measured in time.

Example:

If RPO = **5 minutes**, the organization should be able to recover data from within the last 5 minutes.

Business Continuity Steps

1. Take regular automated backups.
2. Enable data versioning.
3. Replicate important data to another region.
4. Keep backup copies separate from production data.
5. Encrypt backup data.
6. Test restoration regularly.
7. Monitor backup failures.
8. Prepare a disaster recovery plan.
9. Automatically fail over to the backup environment when required.
10. Regularly update and test the recovery procedure.

Testing and Compliance:

11. Run periodic disaster-recovery drills (chaos engineering) to confirm backups and failover actually work as expected.
12. Store backups as immutable copies so ransomware or accidental scripts cannot alter or delete historical versions.
13. Maintain audit logs of backup and restore operations to support compliance reporting for educational data regulations.

Conclusion

Using backup, replication, versioning, and disaster recovery helps the e-learning platform recover quickly from failures and reduces data loss.

4. Hybrid Cloud Architecture for Government Agency

A government agency may have both sensitive and non-sensitive data. Sensitive applications can remain on-premises, while less sensitive applications can be deployed in the public cloud.

Proposed Hybrid Architecture

Users → Secure Gateway → Hybrid Network

On-Premises:

Sensitive databases, confidential applications, and regulated data.

Public Cloud:

Web applications, public information, and less sensitive workloads.

A secure connection such as **VPN or dedicated private connectivity** can connect the on-premises datacenter with the cloud.

Secure Communication

Communication between environments should use:

- Encrypted VPN/private connections.
- Firewalls.
- Network segmentation.
- Secure APIs.
- Intrusion detection and monitoring.

Identity Management

A centralized identity system can be used to manage users across both environments.

Important methods include:

- Single Sign-On (SSO).
- Multi-Factor Authentication (MFA).
- Role-Based Access Control (RBAC).
- Least-privilege access.
- Centralized identity management.

Encryption

Sensitive data should be encrypted:

Data at Rest: Encryption when stored in databases or storage.

Data in Transit: Encryption while data travels between on-premises and cloud systems.

Encryption keys should also be securely managed.

Challenges

Hybrid cloud management can have some challenges:

1. More complex infrastructure.
2. Difficult network management.
3. Different security policies.
4. Identity integration problems.
5. Higher monitoring requirements.
6. Compliance and data-location requirements.
7. Need for skilled cloud professionals.
8. Troubleshooting can be difficult across different environments.

Additional Governance Measures:

9. Adopt a zero-trust approach so every request is verified regardless of whether it originates on-premises or in the cloud.
10. Use policy-as-code to enforce consistent security and compliance rules automatically across both environments.
11. Conduct periodic penetration testing and compliance audits to validate the hybrid boundary remains secure.

Conclusion

A hybrid cloud allows the government agency to keep sensitive workloads on-premises while using the scalability and flexibility of the public cloud for non-sensitive workloads.

5. Multi-Region High Availability for SaaS

A SaaS provider should ensure that its application continues working even if one cloud region fails. A multi-region architecture can provide high availability and fault tolerance.

Proposed Architecture

Users → Global DNS/Traffic Manager → Region 1 / Region 2 / Region 3

Each region can have:

- Application servers.
- Load balancers.
- Databases.
- Storage.
- Monitoring systems.

Multi-Region Deployment

The same application should be deployed in at least two different regions.

For example:

Region 1 → Primary Application

Region 2 → Secondary Application

If Region 1 becomes unavailable, users can be redirected to Region 2.

Data Replication

Data should be replicated between regions.

For example:

Region 1 Database ⇔ Region 2 Database

Depending on application requirements, replication can be synchronous or asynchronous.

- **Synchronous replication:** Data is written to multiple locations before the operation is considered complete. It can provide stronger consistency but may add latency.
- **Asynchronous replication:** Data is copied after the initial write. It can provide better performance but may have some data lag.

Failover

A global traffic manager or DNS service can monitor the health of each region.

If Region 1 fails:

Region 1 → Traffic Manager → Region 2

Users are automatically redirected to the healthy region.

Monitoring and Alerting

Monitoring tools should continuously check:

- CPU and memory.
- Application response time.
- Database health.
- Network performance.
- Error rates.
- Server availability.
- Replication status.

Alerts can be generated when abnormal conditions are detected.

For example:

High Error Rate → Monitoring System → Alert → Cloud Administrator

Benefits

- High availability.
- Reduced downtime.
- Better fault tolerance.
- Improved disaster recovery.
- Better user experience.
- Protection against regional failures.

Cost and Performance Trade-offs:

- Synchronous replication across regions improves consistency but increases latency and infrastructure cost, so it is best reserved for critical transactional data.
- Asynchronous replication is more cost-effective and performant but should be paired with monitoring to track replication lag.
- Right-sizing standby regions (warm standby instead of full active-active) can reduce cost while still meeting availability targets.

Conclusion

A multi-region deployment with data replication, automated failover, load balancing, and continuous monitoring can keep a SaaS application available even when one cloud region fails.

